Mingshi Chen

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EDUCATION

Rice University Aug 2024 - Dec 2025
Master's degree, Electrical and Computer Engineering Houston, Texas, USA

Fudan University Sep 2020 - Jun 2024
Bachelor's degree, Electronic Information Science and Technology Shanghai, China

PUBLICATIONS AND MANUSCRIPTS

DuoGate: Action and Context Control for Reliable Language Agents [.pdf]

Manuscript submitted to ICLR 2027; under review.

DuoGate links action control with selective context adaptation, using execution outcomes and correction feedback to guide learning. Separate gates determine when to learn and which lessons to retain for later decisions.

ADDITIONAL WRITINGS

E-commerce GenAI Assistant I [.pdf] ELEC 594 Capstone Project, Rice University
E-commerce GenAI Assistant II [.pdf] ELEC 594 Capstone Project, Rice University

ADDITIONAL RESEARCH PROJECTS

Workplace Task Generation for Language-Agent Post-Training Sep 2026 - Present

Advised by Pengtao Xie · *Post-training data & evaluation*

- Developing a GDPval-inspired corpus built from O*NET occupations and public workplace documents.
- Tasks pair realistic requests with supporting files, explicit deliverables, and criterion-level evaluation rubrics.
- Implemented a WorkGenesis-based pipeline with source provenance, reproducible occupation schedules, resumable parallel generation, deduplication, bounded retries, and file-integrity checks.
- The broader pipeline supports independent solving, rubric-based judging, defect attribution, and teacher-trajectory collection.

Shared Exploration for Token-Efficient Coding Agents Aug 2026 - Present

Advised by Ning Luo · *Agent systems & efficiency*

- Designed a task-group architecture in which related coding tasks share a persistent exploration agent while keeping planning, implementation, and evaluation separate.
- The goal is to reduce repeated retrieval and context-processing costs.
- Built reproducible task grouping using TF-IDF and cosine similarity, asynchronous evidence exchange, and task-scoped access controls.
- Implemented a CodeProjectEval harness with isolated outputs, official-test scoring, and request-level token accounting that counts shared exploration once per group.

ADDITIONAL RESEARCH PROJECTS CONTINUED

Additional research · Health

Pediatric Brain Tumor Classification Jun 2026 - Present

Advised by Hairong Wang

- Developed a patient-level classification pipeline for whole-slide histopathology using frozen CONCH embeddings and gated-attention multiple-instance learning, incorporating age and anatomical location.
- Audited a cohort of 255 patients and 260 usable slides, and compared hierarchical and global approaches across ten diagnostic classes.
- Improved ten-class accuracy from 56.08% to 63.14% through model refinement and validation-based epoch selection.

Geographic Variation in COPD Care Sep 2026 - Present

Advised by Hairong Wang

- Reproducible analysis of inpatient and outpatient encounter data across 44 Austin ZIP codes and 22 quarters.
- Studied geographic variation in recorded COPD care using observed-to-expected ratios, statistical comparisons, and sensitivity analyses.

Platform-Aware Research Dissemination Assistant Feb 2026 - Jun 2026

Advised by Pengtao Xie

- Built a research dissemination assistant for AIBuildAI, combining public-post retrieval, relevance screening, and platform-specific reply generation for human review.
- Implemented multilingual matching, platform-specific quality checks, and SQLite logging across six platforms.
- Evaluated 91 candidate posts across four platforms; the pipeline retained 40 drafts for review and skipped 51 candidates.

TEACHING

COMP 576 - A Practical Introduction to Deep Machine Learning Fall 2025
Teaching Assistant · Rice University

ELEC 263 - Introduction to Physical Electronics Lab Spring 2025
Teaching Assistant · Rice University

HONORS AND AWARDS

ECE Future Star Scholarship 2024 - 2025
Rice University

First-Class Scholarship 2021 - 2022
Fudan University

ACADEMIC SERVICE

ICML 2026 2026
Conference Reviewer